

Cognome _____ Nome _____ Matricola _____

Basi di Dati

12-01-2015

PROVA SCRITTA

Tot punti: 64

1 (4 pts)	2 (3 pts)	3 (4 pts)	4 (5 pts)	5 (10 pts)	6 (2 pts)	7 (2 pts)	8 (3 pts)	9 (4 pts)	10 (4 pts)	11 (4 pts)	12 (4 pts)	13 (15 pts)

Per la compilazione attenersi alle seguenti istruzioni:

- **Nome e cognome** devono essere scritti in stampatello sul foglio d'esame.
- L'ESAME SI RITIENE CLASSIFICABILE SE: **si ottengono almeno 6 punti negli esercizi: 1, 2, 3**
- Il testo d'esame **deve essere riconsegnato** anche in caso di ritiro dall'esame.
- **È vietato l'uso della matita**, solo le risposte scritte a penna verranno considerate.
- Non sono ammessi appunti o altro materiale, a chiunque copi, comunichi per qualsiasi ragione con i colleghi, o venga trovato ad usare qualsiasi tipo di note o appunti verrà annullato l'esame.
- 60 punti corrispondono al 30/30 (full marks)

1. (4 pts) Given the following CREATE statements for tables :

```
CREATE TABLE Product (  
  pid CHAR(9) PRIMARY KEY,  
  name VARCHAR(250) NOT NULL,  
  description TEXT,  
  price DECIMAL(9,3),  
  prod CHAR(8) NOT NULL,  
  FOREIGN KEY (prod) REFERENCES Producer)
```

```
CREATE TABLE Shipment (  
  pid CHAR(9) REFERENCES Hardware,  
  date TIMESTAMP,  
  quantity INTEGER NOT NULL,  
  PRIMARY KEY (pid, date))
```

```
CREATE TABLE Producer (  
  code CHAR(8) PRIMARY KEY,  
  name VARCHAR(250) NOT NULL,  
  address TEXT)
```

```
CREATE TABLE Hardware (  
  pid CHAR(9) PRIMARY KEY,  
  weight DECIMAL(9,3),  
  fragile boolean,  
  FOREIGN KEY (pid) REFERENCES Product)
```

```
CREATE TABLE Software (  
  pid CHAR(9) PRIMARY KEY,  
  os VARCHAR(10),  
  online boolean,  
  FOREIGN KEY (pid) REFERENCES Product)
```

(a) Design the ER schema that produces the above definitions

(b) Can a Producer produce both Software and Hardware ? ☐ YES ☐ NO ☐ UNDECIDABLE

Non-standard notation (i.e, different from what present in the book and/or in the slides) will be considered a mistake.

DRAW THE ANSWER BELOW:

2. (3 pts) Given the following tables, and with only the tuples present below

$S(A, B, C)$

A	B	C
1	a	a
1	NULL	b
2	a	c
4	a	e

$T(D, E, F)$

D	E	F
a	e	1
b	c	2
c	NULL	2
d	a	3

Taking into consideration only the tuples showed above, without any further assumption, answer the following
[+1 point for correct answers, -1 points for wrong answers, 0 points if answer left blank]

- (a) Can $T(E)$ be foreign key with reference to $S(C)$? ☐ YES ☐ NO ☐ UNDECIDIBLE
- (b) Can (A, C) be a primary key for S ? ☐ YES ☐ NO ☐ UNDECIDIBLE
- (c) Can (D, E) be a primary key for T ? ☐ YES ☐ NO ☐ UNDECIDIBLE

3. (4 pts) Consider the following schema instances (structure and tuples):

$S(A, B)$

A	B
1	10
2	20
3	30

$T(B, C, D)$

B	C	D
1	10	NULL
2	20	f
3	NULL	w

$U(D, E)$

D	E
NULL	f
o	w
f	w
m	NULL

Show the results of the following queries:

WRITE YOUR ANSWER BELOW IN TABLE FORMAT:

(a) `SELECT S.B, T.C +10, T.D
FROM S JOIN T ON S.B < T.C+10`

(b) `SELECT U.E, COUNT(*)
FROM T JOIN U ON T.D<>U.D
GROUP BY U.E`

--

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4. (5 pts) Assume you need to design a database for LEGO® sets. The database stores information about, sets and special editions sets and single components (also called bricks). In addition, the database stores a history of play-dates. Each set has name, a description, and a production code that identifies it. There exist special collection sets, those sets have a date when it has been produced and an additional description and a numeric rating. Each set is composed by many different bricks, special sets are composed by normal bricks and also by special bricks.

Any brick is identified by the shape and color. This means that two bricks with the same shape but different colors are treated as different components. Assume there exists a unique alphanumeric code for shapes. Special edition bricks have an additional description field to describe in which way they are special. A brick can appear in many sets, in different quantities.

A play-date is identified by the friend you meet and the date in which you meet with her to play LEGO®. This means that if you meet two different friends in the same date, you will register two separate play-dates. During a play-date you register which sets you have built.

Provide the following:

- (a) the ER diagram for the database to respect all above conditions.
- (b) the relational schema of the database (mark also foreign keys in the schema)

Non-standard notation (i.e, different from what present in the book and/or in the slides) will be considered a mistake.

DRAW THE ANSWER BELOW:

5. (10 pts) A database stores information about product purchases and is designed with the following schema:

Product (code, name, price, weight, discontinued, supplier, address)

Cart (customer, date, prod_code, prod_name, quantity, price, suppl, cust_name, cust_addr)

The product is identified by a code. Then the database stores the name, price, weight and a boolean flag to mark it as discontinued. It also stores the code of the supplier, and its address.

The cart refers to a set of products bought together. It register when a customer buys different products in the same date, i.e., the same cart. It refers to the customer id, and the product code. Then it stores the product name, the quantity of product bought, the price of the single item, the code of the supplier, and the name and address of the customer doing the purchase. If two different products are bought in the same date, by the same customer, are considered in the same cart.

Each product has a single supplier. Each supplier has a single address. Each customer has a single name and a single address.

(a) Provide in **RELATIONAL ALGEBRA** the query to return suppliers with no discontinued products.

(b) Provide in **RELATIONAL ALGEBRA** the query to return the most expensive product (or products).

(c) Provide in **SQL** the query to find customers who bought all products.

(d) Provide in **SQL** the query to find for all suppliers the average price of all their products.

(e) Provide in **SQL** the query to find for every customer the most expensive purchase: an entry in **Cart** with the largest price multiplied by quantity of product bought.

6. (2 pts) Given the Relational Schema of Exercise 5:

Product (code, name, price, weight, discontinued, supplier, address)

Cart (customer, date, prod_code, prodname, quantity, price, suppl, cust_name, cust_addr)

with the same assumptions described above

(a) Find all the functional dependencies

(b) Compute the BCNF decomposition

7. (2 pts) Date le circostanze $R(A, B, C, D, E)$ e $S(F, G, H)$, dire per ognuna delle seguenti dipendenze funzionali, se sono valide per l'istanza delle tabelle sotto riportate.

[+0.5 per risposta corretta, -0.5 per ogni risposta sbagliata, 0 per ogni risposta non data]

A	B	C	D	E
1	4	1	3	6
2	4	2	4	7
2	4	2	5	7
2	4	2	6	7

F	G	H
1	1	1

$A \rightarrow CD$ ☐ SI ☐ NO

$AD \rightarrow E$ ☐ SI ☐ NO

$A \rightarrow E$ ☐ SI ☐ NO

$F \rightarrow GH$ ☐ SI ☐ NO

8. (3 pts) Answer the following questions

[+1 point for correct answers, -1 points for wrong answers, 0 points if answer left blank]

- (a) If R and S are relations with only common attribute a , then it's always true that $\sigma_{R.a=3}(R \times S) \equiv \sigma_{R.a=3}(S \times R)$.
☐ YES ☐ NO ☐ UNDECIDABLE.
- (b) Attribute A for a relation R can be Foreign Key referencing B in relation S , even if A is not primary key for R .
☐ YES ☐ NO ☐ UNDECIDABLE.
- (c) The $\bowtie$ operator in relational Algebra can be equivalently expressed by using $\cap$ and σ operators
☐ YES ☐ NO ☐ UNDECIDABLE.

9. (4 pts) Answer the following questions

[+1 point for correct answers, -1 points for wrong answers, 0 points if answer left blank]

- (a) The cost of $\sigma_{A=1}(R \bowtie S)$ is the same as $\sigma_{A=1}(R) \bowtie S$ if the attribute A occurs both in R and S .
☐ TRUE ☐ FALSE ☐ UNDECIDABLE.
- (b) If attributes A and B of relation R form a foreign key that is referencing attributes C and D of relation S , then the attribute A cannot be null
☐ TRUE ☐ FALSE ☐ UNDECIDABLE.
- (c) If attributes A and B of relation R form a foreign key that is referencing attributes C and D of relation S , then the attribute A has to be a key
☐ TRUE ☐ FALSE ☐ UNDECIDABLE.
- (d) The maximum number of unique statements someone can define on a relation of 5 columns is
☐ 0 ☐ 1 ☐ 5 ☐ 5^2 ☐ 2^5 ☐ $\log 5$ ☐ 5^5 ☐ INFINITE ☐ NONE OF THOSE.

10. (3 pts) Consider a relation $R[A, B, C, D, E]$ and the FDs $A \rightarrow BC$, $BD \rightarrow C$. Is it in BCNF ? Can you provide a BCNF dependency preserving decomposition? If yes, provide it. If not, explain why.

11. (3 pts) Consider the following schema:

Professor(name, office, phone)

Course(course#, semester, instructor_name, enrolment)

that is, for every course of every semester, the name of the instructor who taught it and the enrolment (meaning how many students were in the class)).

The instructor_name is a FK to the Professor.name.

Enroll(student_id, course#, semester)

Write in words in one sentence for a person that does not know SQL what the following query finds:

select name from Professors

minus

select P.name from Professors P, Course C where P.name=C.instructor_name

12. (3 pts) Consider the following schema:

Person [Pid, Name, Salary, Lives]

Building [Bid, Expenses, Amministratore]

Person.Lives is foreign key referencing Building.Bid and Building.Amministratore is foreign key referencing Person.Pid

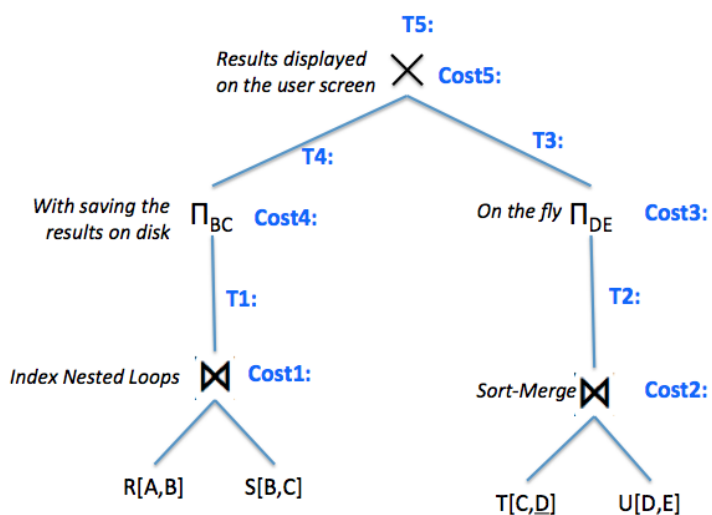
If the most common query is the one that finds the Pid of the person with the highest salary, what is the best index (or best indexes) to create in order to make this fast?

13. (10 pts) Consider the 4 relations $R[A, B]$, $S[B, C]$, $T[C, D]$, $U[D, E]$ and you know the following information about each one of them:

$R[A, B]$	$S[B, C]$	$T[C, D]$	$U[D, E]$
has a B+tree on B	has P_S pages	has P_T pages	has P_U pages
has P_R pages	A tuples has size T_s	has C_T number of tuples	has C_U number of tuples
B has selectivity factor f_B	50% of B is null	T.D is a key	D has no null values
has C_R number of tuples			

You also know that a page on the disk is of size P and any possible operators makes no duplicate elimination. Furthermore, the memory has 3 buffers that each one is as big as a page of the disk. A B+tree lookup costs 3.

Based on that information write the formula that computes the cost of each operators (Cost1, Cost2, Cost3 and Cost4) and for each operator write the number of tuples that the operator generates (T1, T2, T3 and T4). In the formula you can use any of the values mentioned in the table above and any of the variables Cost1, Cost2, Cost3, Cost4, T1, T2, T3, and T4.



Cost1=

T1=

Cost2=

T2=

Cost3=

T3=

Cost4=

T4=

Cost5=

T5=

Spazio Libero

Spazio Libero